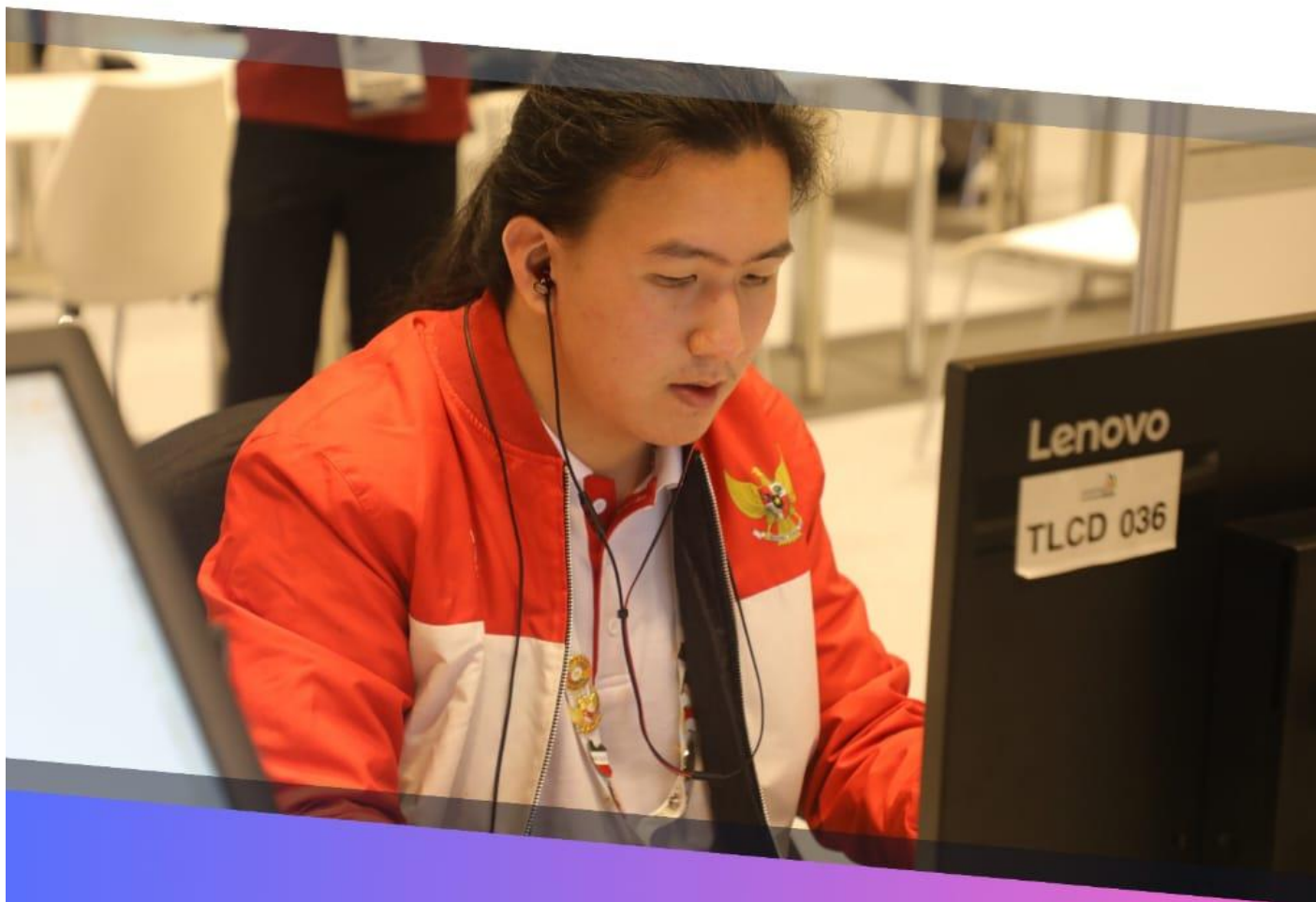




TAHUN
2024

TEST PROJECT

SELEKNAS CALON KOMPETITOR THE 14th WORLDSKILLS ASEAN IT SOFTWARE SOLUTIONS FOR BUSINESS



**KEMENTERIAN KETENAGAKERJAAN R.I.
DIREKTORAT JENDERAL PEMBINAAN
PELATIHAN VOKASI DAN PRODUKTIVITAS
DIREKTORAT BINA STANDARDISASI KOMPETENSI DAN
PROGRAM PELATIHAN**

Gedung Vokasi, Jl. Jend Gatot Subroto Kav. 44
Jakarta Selatan 12710

<https://inaskills.kemnaker.go.id>

Motorcycle Repair Application

General Guidelines

- 1) For this module, you must **use the given sql file to run DDL and DML for creating database used for this module**. Please check the sql folder given inside the test project. **You must not modify structure of the database and tables**. However, **you are allowed to modify (such as insert, update, or delete) the records**. However please note that marking process will not be using your modified database.
- 2) You must submit **the whole C# Project**. Pay attention to the **local resource files** you may add during your works (for example image files) if it needs to be included in your project submission.
- 3) Make sure your project **capable to run even in different environment setup**. During marking process, we may as far as try to modify your **string connection** as needed. After that, if we found that your **project cannot be compiled, the zero mark will be given** instead.
- 4) Create a project with the name '**Motor[Your Last Name]**'.

A motorcycle repair shop wants to create an application that can manage user data that can access the application, services and products provided, mechanics who handle motorcycle services, and add motorcycle service transactions. Your task now is to **create the prototype using c# programming language**. The **database diagram** can be seen as **figure 1**. Details of the application are as follows.

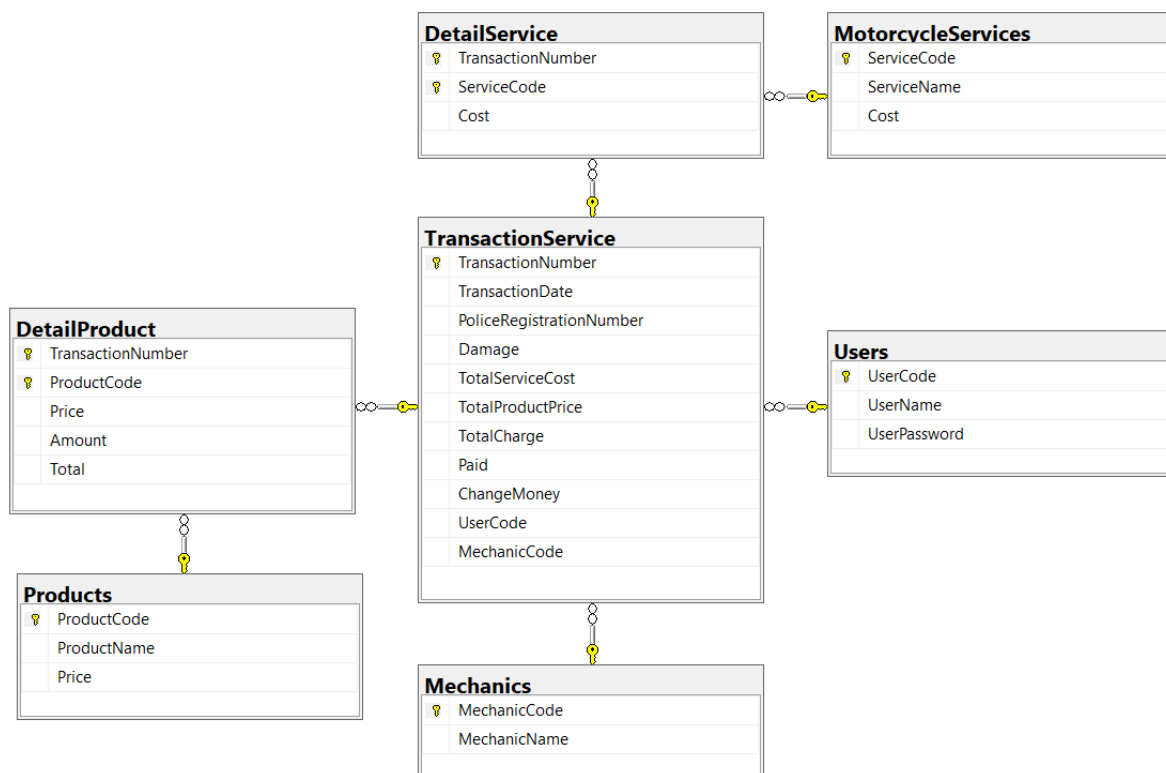


Figure 1. Database Diagram

A. Login Form

- This form can be seen as Figure 2.
- This form is displayed in the center of the screen.
- User code is filled with user_code and password is filled with password_user from the users table.
- If the user enters an incorrect user code, a notification will appear saying "user not found".
- If the user enters the correct user code but the password is incorrect, a notification will appear saying " incorrect password".
- After successful login, they will be directed to the Main Menu Form (Figure 3).

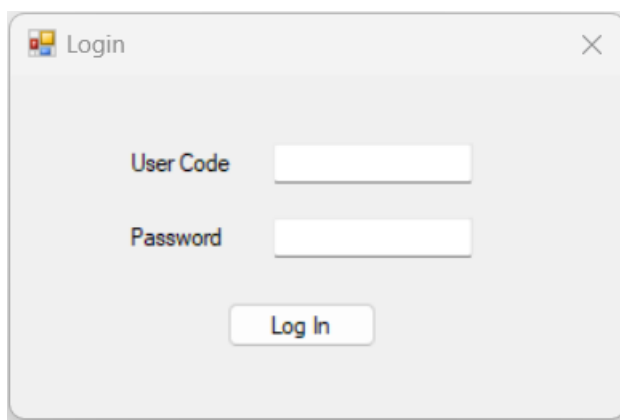


Figure 2. Login Form

B. Main Menu Form

- This form can be seen as Figure 3.
- This form is displayed in maximized state.
- There are three menus on this form, that is Master, Transaction, and Log Out.
- In the Master menu there are four submenus, that is Users, Products, Services, Mechanics.
- Users submenu will be directed to the Users Form (Figure 4).
- Products submenu will be directed to the Products Form (Figure 5).
- Services submenu will be directed to the Services Form (Figure 6).
- Mechanics submenu will be directed to the Mechanics Form (Figure 7).
- Transaction menu will be directed to the Transaction Form (Figure 8).
- Log Out menu to exit the application.

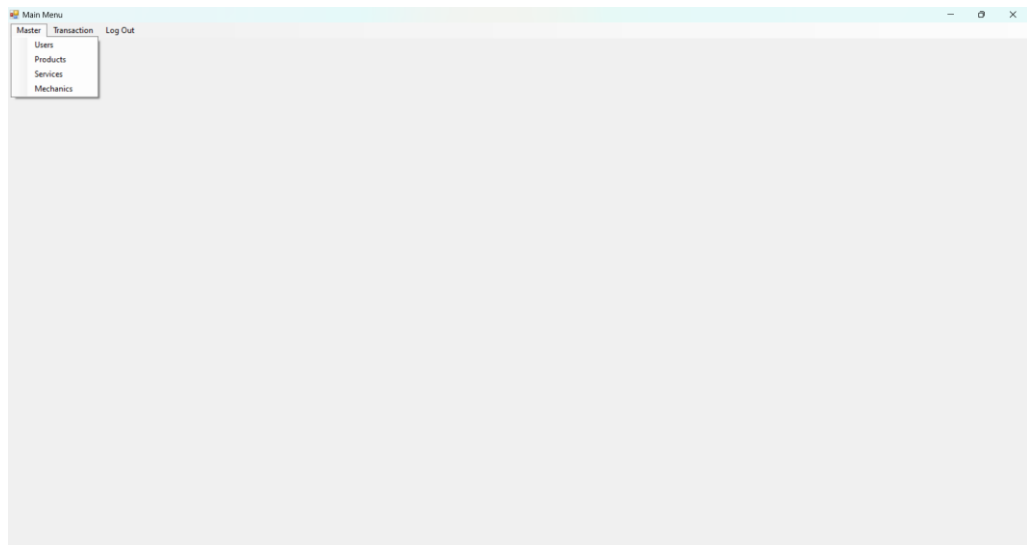


Figure 3. Main Menu Form

C. Users Form

- This form can be seen as Figure 4.
- This form is displayed in the center of the screen.
- Datagridview displays all data from the users table with a layout format as shown in Figure 4.
- User Code must be autogenerated with the format 'USR-[two digits number of the current year (YY)]-[two digits user sequence number]' and cannot be filled in manually.
- Validate name of the user (Name of the user minimum 3 characters and maximum 25 characters).
- Add data into the database by fill all the required fields and then click the save button.
- Edit data in the database by clicking the data row in the datagridview then the clicked data will be displayed in the textbox. Change the data in the textbox as needed then click the edit button.
- Delete data in the database by clicking the data row in the datagridview then the clicked data will be displayed in the textbox. Check the data to be deleted in the textbox then click the delete button.
- The cancel button is used to clear data in all textbox and re-generate the user code.

	User Code	Name	Password
▶	USR-20-01	Agus Supriadi	y7IGGjwH
	USR-20-02	Denira Fitria	tTRBiju9
	USR-22-01	Jessica Andriani	iXEyVoLc
	USR-23-01	Clara Rika	5ycAXFEK
	USR-24-01	Puspa Winarsih	MfY5RCrM
*			

Figure 4. Users Form

D. Products Form

- This form can be seen as Figure 5.
- This form is displayed in the center of the screen.
- Product Code must be autogenerated with the format 'PR[three digits product sequence number]' and cannot be filled in manually.
- Price textbox can only be filled with numbers.
- Datagridview displays all data from the products table with a layout format as shown in Figure 5.
- Add data into the database by fill all the required fields and then click the save button.
- Edit data in the database by clicking the data row in the datagridview then the clicked data will be displayed in the textbox. Change the data in the textbox as needed then click the edit button.
- Delete data in the database by clicking the data row in the datagridview then the clicked data will be displayed in the textbox. Check the data to be deleted in the textbox then click the delete button.
- The cancel button is used to clear data in all textbox and re-generate the product code.

	Product Code	Product Name	Price
▶	PR001	ACCELERATOR GRIP	30000
	PR002	ACCELERATOR / THROTTLE CABLE	30000
	PR003	AIR CLEANER COMPLETE	120000
	PR004	AIR CLEANER	25000

Figure 5. Products Form

E. Services Form

- This form can be seen as Figure 6.
- This form is displayed in the center of the screen.
- Service Code must be autogenerated with the format 'SR[three digits service sequence number]' and cannot be filled in manually.
- Cost textbox can only be filled with numbers.
- Datagridview displays all data from the MotorcycleServices table with a layout format as shown in Figure 6.
- Add data into the database by fill all the required fields and then click the save button.
- Edit data in the database by clicking the data row in the datagridview then the clicked data will be displayed in the textbox. Change the data in the textbox as needed then click the edit button.
- Delete data in the database by clicking the data row in the datagridview then the clicked data will be displayed in the textbox. Check the data to be deleted in the textbox then click the delete button.
- The cancel button is used to clear data in all textbox and re-generate the service code.

Service Code	Service Name	Cost
SR001	Service Matic 110cc	95000
SR002	Service Matic 125cc	110000
SR003	Service Matic 150cc	120000
SR004	Service Manual 110cc	125000

Figure 6. Services Form

F. Mechanics Form

- This form can be seen as Figure 7.
- This form is displayed in the center of the screen.
- Mechanic Code must be autogenerated with the format 'MC[three digits service sequence number]' and cannot be filled in manually.
- Datagridview displays all data from the mechanics table with a layout format as shown in Figure 7.
- Add data into the database by fill all the required fields and then click the save button.
- Edit data in the database by clicking the data row in the datagridview then the clicked data will be displayed in the textbox. Change the data in the textbox as needed then click the edit button.
- Delete data in the database by clicking the data row in the datagridview then the clicked data will be displayed in the textbox. Check the data to be deleted in the textbox then click the delete button.
- The cancel button is used to clear data in all textbox and re-generate the service code.

	Mechanic Code	Mechanic Name
▶	MC001	Toni Setiawan
	MC002	Ardi Mandala
	MC003	Eman Megantara
	MC004	Lutfan Wibowo
	MC005	Rangga Prakasa
	MC006	Ajiono Halim
	MC007	Catit Wulandari

Figure 7. Mechanics Form

G. Transaction Form

- This form can be seen as Figure 8.
- This form is displayed in the center of the screen.
- Transaction Number must be autogenerated with the format 'T[three digits service sequence number of the month][four digits of the current month and year (MMYYYY)]' and cannot be filled in manually.
- Date must be autogenerated with a layout format as shown in Figure 8.
- Police Registration Number and Damage textbox value typed by user.
- Add service to datagridview service by typing the service code and press enter, then the service name and cost will appear on the datagridview service according to the service code typed. Total Service, Total Service Cost, and Total Charge are automatically updated every time a service is added to the datagridview.
- Delete a service on the datagridview service by double clicking on the data row of the datagridview service to be deleted, the deleted data disappears from the datagridview service. Total Service, Total Service Cost, and Total Charge are automatically updated every time a service is deleted on the datagridview.
- Add product to datagridview product by typing the product code and press enter, then the product name, price, amount (default 1), and total will appear on the datagridview service according to the service code typed. Users can change the amount value of the product by typing in the amount column, then the total column automatically changes according to the price and amount. Total

Product, Total Product Price, and Total Charge are automatically updated every time a product is added to the datagridview or user changes the amount.

- Delete a product on the datagridview product by double clicking on the data row of the datagridview product to be deleted, the deleted data disappears from the datagridview product. Total Product, Total Product Price, and Total Charge are automatically updated every time a service is deleted on the datagridview.
- The Mechanic dropdown/combo box displays the Mechanic Name (not Mechanic Code).
- The Paid textbox is typed by the user whose value cannot be less than the Total Charge textbox value.
- Textbox change automatically filled with paid value - total charge value.
- The save button is used to save the transaction form data that has been filled in by the user into the database (TransactionService table, DetailService table, DetailProduct table). After the data is successfully saved, re-generate the transaction number and date then clear the previous user input on the transaction form.
- The cancel button is used to re-generate the transaction number and date also clear the user input on the transaction form.

Transaction

Motorcycles

Transaction Number
T001112024
Police Registration Number
B2010SYT

Date
04 Nov 2024
Damage
Thin front tire

Service

	Service Code	Service Name	Cost
	SR005	Service Manual 125cc	135000
	SR007	Tire Replacement Service	15000
▶*			

Total Service
2
Total Service Cost
150000

Product

	Product Code	Product Name	Price	Amount	Total
	PR012	OIL FILTER	85000	1	85000
	PR019	FRONT TYRE	360000	1	360000
	PR021	REARVIEW MIRROR	45000	2	90000

Total Product
4
Total Product Price
535000

Mechanic

Mechanic Name
Ardi Mandala

Process

Save
Cancel

Charge

Total Charge
685000
Paid
700000
Change
15000

Figure 8. Transaction Form